

Digital Payments and Formal Borrowing Across 97 Economies: A Pre-registered Test of the Credit-Information-Substitution Hypothesis

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Abstract

We ask whether digital-payment activity is associated with formal borrowing among adults, and whether that association is larger where credit-information coverage is thinner — the individual-level analogue of the firm-level pattern documented by Galilea et al. (2026). Using the 2024 Global Findex microdata for 100,560 adults in 97 economies, we estimate survey-weighted linear probability models with economy fixed effects and economy-clustered standard errors, following a pre-registered plan. Making or receiving a digital payment is associated with a 3.617 percentage-point-higher probability of having borrowed from a formal financial institution in the past 12 months (95% CI 2.597–4.638 pp), conditional on account ownership, demographics and economy fixed effects. The pre-registered moderation prediction is **not supported**: the interaction of digital-payment activity with *lower* 2019 credit-information coverage is negative (-1.975 pp per economy-level standard deviation; 95% CI -2.916 to -1.033 pp), the opposite of the hypothesised sign. For the registered coverage moderator, the negative interaction persists across weighting, sample and control-set changes and all 97 leave-one-economy-out fits. The observed clustered precision gives an approximate 80%-power minimum detectable interaction of 1.33 pp per SD. Two alternative institutional proxies are imprecise and do not reproduce a significant reversal. The registered receive-versus-make mechanism and the informal-borrowing placebo also fail. We interpret the results as a cautious, non-causal external-population check that does not reproduce the credit-information-substitution reading at the individual level, and we discuss why a coverage floor and the account-access channel are plausible explanations. We report Vietnam’s position in the cross-country distribution for local context.

Keywords: digital payments, financial inclusion, formal borrowing, credit information, Global Findex, pre-registration. **JEL:** G21, G51, O16, D14.

1 Introduction

Digital payments have become widely used in low- and middle-income economies: the 2024 Global Findex shows that a majority of adults make or receive at least one digital payment (Klapper et al., 2025). A recurring policy claim is that the transaction record such payments generate can serve as an alternative credit signal, extending formal credit to borrowers whom

traditional credit-information systems do not cover. Galilea et al. (2026) provide the most direct evidence for this idea: across 101 economies, firms that use electronic payments report weaker credit constraints, and the association is stronger where credit-information depth is lower — consistent with electronic-payment records substituting for a missing credit file.

Whether the same pattern generalises to *individuals* remains an open empirical question. Our targeted, non-systematic search found no prior multi-country individual-level test of the exact Findex interaction, but cannot establish priority. Individuals face different lenders, different products and different information technologies than firms, and the Global Findex — the natural data source at the individual level — has been used mainly to describe digital-payment adoption and broad financial-inclusion outcomes rather than to test this specific moderation. Berg et al. (2020) show that digital-footprint variables can predict individual default and add information to a credit-bureau score. Their complementarity result is a competing explanation, not evidence for our pre-registered substitution prediction, and their setting is a single e-commerce lender in one country. Bazarbash and Beaton (2020) relate cross-country fintech credit to financial development and information infrastructure using provider-level data, again not at the individual level.

This paper runs a pre-registered individual-level test. We do **not** claim novelty of the question or of the method; we position the exercise as an individual-level descriptive analogue and an external-population check of the established firm-level pattern. Three features discipline the exercise. First, the full analysis plan — sample, variables, specifications, robustness checks, multiple-testing correction and decision rules — was frozen and hash-committed before any outcome model was estimated. Second, the primary results were produced by one implementation and then independently reproduced by a separate cold implementation that did not see the first; 41 of 44 compared quantities match to better than 5×10^{-13} . Third, the decision rules were written to make a null or contrary interaction a valid, reportable outcome rather than a failure.

The pre-registration matters because the headline result is a contrary one. We find that digital-payment activity is positively associated with formal borrowing within economies (3.617 pp, 95% CI 2.597–4.638), but that this association is **smaller**, not larger, in economies where 2019 credit-bureau and credit-registry coverage was lower (interaction -1.975 pp per standard deviation, 95% CI -2.916 to -1.033). The sign reversal is robust to unweighted and population weights, to excluding high-income economies, to adding country-level macro controls and their interactions, to dropping zero-coverage economies, to leaving out each of seven world regions, and to leaving out each of the 97 economies one at a time. The depth and legal-rights proxies are statistically imprecise and do not establish a general institutional gradient. The registered mechanism split (receiving versus making payments) and the informal-borrowing placebo also do not behave as the substitution reading predicts.

Our contribution is therefore an honest, reproducible negative result. At the individual level, in this cross-section, the cross-country pattern that motivates the digital-footprint-substitution policy narrative does not appear; if anything the data show the opposite gradient. We evaluate plausible explanations — a coverage floor in the thinnest-coverage economies, and the possibility that in those economies digital-payment activity mainly proxies for basic account access rather than for an incremental credit signal — and we are explicit about what the design cannot rule

out.

2 Related literature and positioning

Firm-level evidence. Galilea et al. (2026) is the closest paper. Using World Bank Enterprise Survey data for 101 economies, they document that electronic-payment use is associated with weaker firm credit constraints and that the association is larger where the depth-of-credit-information index is lower. Our design mirrors theirs at the individual level: same broad hypothesis, same moderation logic, different unit and different data. Their robustness section changes firm-level outcomes, payment instruments, fixed effects and controls within the Enterprise Surveys; neither the paper nor its robustness appendix uses Global Findex or estimates an individual-level borrowing specification. Thus it sharpens, rather than duplicates, the external-population test conducted here.

Digital footprints and credit. Berg et al. (2020) show that easily accessible digital-footprint variables match or exceed a credit bureau score in predicting consumer default at a German e-commerce lender and, crucially, complement rather than replace bureau information. This establishes that the *information content* of a digital trace can be real; it does not establish that the trace expands formal borrowing at the population level, still less that the effect concentrates where bureaus are thin.

Cross-country digital credit. Bazarbash and Beaton (2020) find a market-level gap-filling pattern, while Cornelli et al. (2023) conclude that fintech and big-tech credit complement rather than substitute for bank credit. Both study aggregate provider volumes rather than an individual borrower’s use of formal credit. The disagreement suggests that substitution and complementarity can depend on the level of analysis and the lending channel.

Closest individual-level evidence. Alok et al. (2024) study India’s UPI rollout with a causal design and find credit growth concentrated in places where account access was more recent; only fintech lenders expanded lending to new-to-credit borrowers. Their within-country setting, treatment and outcomes differ sharply from ours, but the result makes basic formal-system access a serious competing explanation for the cross-country gradient.

Global Findex. Klapper et al. (2025) and the associated database document the rapid adoption of digital payments and its correlation with account ownership and with saving and borrowing behaviour. The database is descriptive by construction. We use it for a specific pre-registered moderation test rather than for adoption description.

Positioning. We claim only an individual-level descriptive analogue and external-population check of the Galilea et al. pattern. A targeted search found no prior individual-level, multi-country Findex study of this exact interaction, but search coverage is incomplete and does not justify a categorical priority claim. We make no causal or mechanism-identification claim; novelty remains a candidate distinction rather than an established fact.

3 Data

3.1 Individual data

The individual data are the 2024 wave of the World Bank Global Findex microdata (<https://microdata.worldbank.org/catalog/7860>; DOI 10.48529/bk9n-8r43; SHA-256 ca307a0c...f90fce). The borrowing and digital-payment module was administered in 98 of the 140 released economies; the reduced questionnaire elsewhere omits the outcome. Our unit of observation is an adult aged 15 or older within an economy.

- **Outcome formal_borrow**: one if the respondent reports having borrowed from a bank or another type of formal financial institution in the past 12 months ($fin22a = 1$), zero if not ($fin22a = 2$); “don’t know” and “refused” are dropped.
- **Exposure anydigpayment**: one if the respondent made or received at least one digital payment in the past 12 months, zero otherwise.
- **Mechanism split** (secondary): **receive_dig**, one if the respondent received any of wages, government transfers, a pension or an agricultural payment into an account; **make_dig**, one if the respondent made a digital merchant payment or paid a utility bill from an account. Unknown responses are treated as missing, never as zero, and are dropped only for the mechanism regressions.
- **Controls** (frozen primary set): account ownership at a financial institution, an indicator for female, age centred and its square, three education categories, five within-economy income quintiles, and an urban indicator.

3.2 Country moderator

The moderator is the country’s **2019 credit-information coverage**: the maximum of the 2019 private credit-bureau coverage and the 2019 public credit-registry coverage, in percent of adults, from the archived World Bank Doing Business series (IC.CRED.ACC.PRVT.CRD.ZS, IC.CRED.ACC.PUBL.CRD.REG.COVR.ZS; snapshot SHA-256 ef0c67f9...bd06c). We denote its negative z -score, computed across the estimation economies, **lowcov2019_z**, so that higher values mean thinner 2019 coverage. Across the 97 estimation economies the raw coverage measure has a cross-economy (unweighted) mean of 41.5%, a median of 40.4%, and a wide spread (24 economies below 10%). The depth-of-credit-information index and the strength-of-legal-rights index, both 2019 vintage, are used as distinct robustness proxies, not as the same construct.

The moderator is measured five years before the outcome. Credit infrastructure is slow-moving, so this is a defensible historical measure, but we treat the vintage mismatch as a limitation and never describe the moderator as the contemporaneous 2024 system.

The source also requires an integrity qualification. The World Bank paused and later discontinued *Doing Business* after reviews of irregular data changes in the 2018 and 2020 report cycles. The Bank’s December 2020 review specifically identified *Getting Credit* irregularities for China (DB2018) and Saudi Arabia (DB2020), both present in our estimation sample. Our moderator uses archived 2019-vintage coverage fields rather than an overall rank, so the review

does not by itself establish that every value used here is erroneous. We nevertheless treat the history as a measurement-risk warning and report a sensitivity that excludes both economies.

As a partial external check, the 2019 bureau- and registry-coverage, depth and legal-rights values were cross-validated against the published *Doing Business 2020* economy profiles for eight contrasting economies (Vietnam, Afghanistan, Kenya, Georgia, China, South Africa, Saudi Arabia and Panama), and Vietnam’s 2019 bureau coverage was additionally confirmed against a live World Bank API query; all matched the archived snapshot, including the extreme values (zero-depth Afghanistan, full-coverage Georgia and China). This does not rule out error in the economies not checked, and the broad measurement-risk limitation stands.

3.3 Country controls (robustness only)

For one robustness specification we add, at the economy level, log GDP per capita and domestic credit to the private sector (% of GDP), each the latest World Bank observation no later than 2023, and the in-sample economy account-ownership rate, together with their interactions with digital-payment activity. GDP per capita is 2023 for every economy; the private-credit series has heterogeneous vintages (2008–2023), which we disclose.

3.4 Sample construction

Table 1 records the deterministic sample. Requiring a valid formal-borrowing response leaves 102,644 adults in 98 economies. Requiring complete primary covariates removes a further 2,084 adults and one economy (Lesotho, whose income-quintile variable is entirely missing). The moderator merges to every remaining economy by ISO3 code. The estimation sample is **100,560 adults in 97 economies**; the mechanism sub-sample, which additionally requires non-missing receive and make indicators, is 90,118 adults in 89 economies. Vietnam contributes 998 respondents.

Table 1: Sample construction.

Step	Adults	Economies
Valid formal-borrowing response ($fin22a \in \{1, 2\}$)	102,644	98 (42 economies exit)
+ non-missing digital-payment indicator	102,644	98
+ non-missing account ownership	102,644	98
+ non-missing sex	102,644	98
+ non-missing age	102,644	98
+ non-missing education	102,245	98
+ non-missing income quintile	101,247	97 (LSO exits)
+ non-missing urbanicity	100,560	97
+ 2019 coverage moderator merged by ISO3	100,560	97

Notes. Deterministic listwise construction from the 2024 Global Findex microdata (catalogue 7860). “Don’t know” and “refused” on the outcome are dropped before step 1. The moderator merges to every remaining economy.

Raw rates (survey-weighted): 3.78% of adults with no digital payment have borrowed formally, against 14.13% of adults with a digital payment — an unadjusted gap of about 10.3 percentage points, which the regressions below reduce to 3.617 pp once account ownership,

demographics and economy fixed effects are held fixed.

The 97-economy frame spans all seven World Bank regions represented by the module: 34 economies are in Sub-Saharan Africa, 17 in Europe and Central Asia, 16 in Latin America and the Caribbean, 10 in the Middle East and North Africa, 9 in East Asia and the Pacific, 5 in South Asia, and 6 are classified as high income. The 24 economies with coverage below 10% contribute 23,458 respondents, or 23.3% of the estimation rows. These are unweighted sample counts; the primary estimator gives every economy equal total weight.

Table 2: Descriptive statistics.

<i>Panel A. Individual characteristics (survey-weighted means)</i>				
	All ($N = 100,560$)	No digital payment ($N = 44,368$)	Digital payment ($N = 56,192$)	
Formal borrowing	0.092	0.038	0.141	
Account at a financial institution	0.505	0.182	0.805	
Female	0.512	0.558	0.469	
Age (years)	38.6	38.2	39.1	
Urban	0.390	0.339	0.438	
<i>Panel B. Economy-level 2019 moderator ($G = 97$ economies, unweighted)</i>				
	Mean	Median	[p25, p75]	Vietnam
2019 credit-information coverage (%)	41.5	40.4	[10.4, 59.9]	59.4
2019 private-bureau coverage (%)	32.2	20.6	[0.0, 56.5]	20.6
2019 public-registry coverage (%)	16.0	2.1	[0.0, 26.1]	59.4
Depth-of-credit-information index (0–8)	5.9	7.0	[6.0, 8.0]	8.0
Strength-of-legal-rights index (0–12)	6.0	6.0	[3.0, 8.0]	8.0

Notes. Panel A: means weighted by `w_equal` (each economy sums to one). Panel B: one row per economy, unweighted; the moderator used in estimation is the negative z -score of the coverage measure (`lowcov2019_z`).

4 Empirical strategy

All specifications were frozen in the pre-registration and hash-committed before estimation.

4.1 Primary specification

For adult i in economy c , let y_{ic} be formal borrowing, D_{ic} digital-payment activity, A_{ic} account ownership, and L_c the lower-2019-coverage moderator (`lowcov2019_z`). We estimate, by weighted least squares,

$$y_{ic} = \beta_1 D_{ic} + \beta_2 A_{ic} + X'_{ic} \gamma + \delta_c + \varepsilon_{ic} \quad (\text{M1})$$

and, for the moderation test,

$$y_{ic} = \beta_1 D_{ic} + \beta_3 (D_{ic} \times L_c) + \beta_2 A_{ic} + \beta_4 (A_{ic} \times L_c) + X'_{ic} \gamma + \delta_c + \varepsilon_{ic}. \quad (\text{M2})$$

Here X is the frozen control set, δ_c are economy fixed effects (which absorb the moderator's main effect), and weights `w_equal` make each economy sum to one, so the estimand is the average within-economy association with economies weighted equally. Standard errors are clustered by economy (97 clusters); we also report Rademacher wild-cluster-bootstrap p -values (999

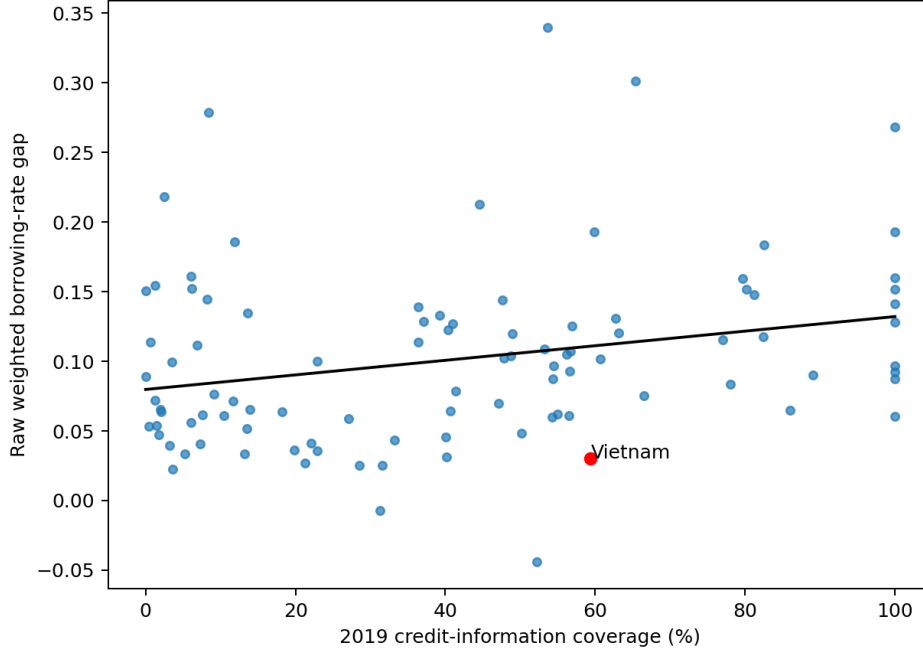


Figure 1: Economy-level raw (survey-weighted) gap in formal borrowing between adults with and without a digital payment, against 2019 credit-information coverage. One point per economy; the fitted line is a simple OLS through the economy points; Vietnam is marked. Source: 2024 Global Findex; 97 economies.

replications). β_3 is the **pre-registered test**. $\beta_3 > 0$ would mean the digital-payment/formal-borrowing association is stronger where 2019 coverage was thinner.

4.2 Pre-registered decision rule

The frozen rule: H-000500 is *supported* only if $\beta_3 > 0$, significant at 5%, and robust to the below-median moderator contrast, the bureau/registry joint test, alternative weighting, dropping high-income economies, the depth and legal-rights proxies, and country-level controls, and if the mechanism split shows the receive interaction positive. $\beta_3 \approx 0$ or $\beta_3 < 0$ means *not supported*, to be reported plainly. A non-null placebo interaction for informal borrowing also undermines the substitution reading.

4.3 Secondary and robustness specifications

The plan specifies: a below-median moderator contrast (M2b); a bureau/registry joint test (M2c); the receive-versus-make mechanism split (M3); three secondary outcomes with Benjamini–Hochberg control over a frozen nine-hypothesis family (M4); four heterogeneity interactions (M5); a ten-branch robustness grid (M6, including unweighted and population weights, dummy-variable logit/probit, dropping high-income economies, the depth and legal-rights proxies, country controls, a minimum economy-size filter, an internet-use control, an economy-block bootstrap, a leave-one-economy-out sweep and an employment control); and a descriptive Vietnam-positioning exercise (M7). Adversarial checks (drop each control, restrict to account holders, drop zero-coverage economies, drop the lowest coverage quartile, leave out each region,

and a moderator sign-reversal algebra check) were run after estimation.

4.4 Interpretation limits

The design is a single cross-section. Economy fixed effects remove all fixed cross-country confounding but do not identify the effect of digital-payment use or of the moderator, and do not make the cross-level interaction causal. Digital-payment activity and formal borrowing are mechanically linked through account ownership, which we condition on but which is itself a possible common consequence. We therefore use associational language throughout and prohibit causal, policy-effect and mechanism-identification claims.

5 Results

5.1 The association

Digital-payment activity is associated with a **3.617 percentage-point** higher probability of formal borrowing (M1; economy-clustered SE 0.514 pp; $p \approx 3 \times 10^{-10}$; 95% CI 2.597–4.638 pp; wild-cluster-bootstrap $p = 0.001$; 100,560 adults, 97 economies). Against a sample formal-borrowing rate of about 9.2%, this is an association of roughly 40% of the mean. It is a within-economy, covariate-adjusted association, not a causal effect. Table 3 reports M1 and M2 in full.

Table 3: Digital payments, formal borrowing and the 2019 coverage moderator.

	(1) M1 Association	(2) M2 Moderation
Digital payment (β_1)	3.617*** (0.514)	3.957*** (0.482)
Digital payment $\times$ lower 2019 coverage (β_3)	– –	–1.975*** (0.474)
Account at FI (β_2)	8.182*** (0.723)	7.800*** (0.659)
Account at FI $\times$ lower 2019 coverage (β_4)	– –	1.828*** (0.616)
Demographic controls	Yes	Yes
Economy fixed effects	Yes	Yes
Clusters (economies)	97	97
Observations	100,560	100,560
Within- R^2	0.0475	0.0481

Notes. Survey-weighted (`w_equal`) linear probability models; coefficients and standard errors in percentage points. Economy-clustered (CR1) standard errors in parentheses. Demographic controls: female, age (centred) and its square, education (3 categories), within-economy income quintile (5), urban. β_3 is the pre-registered test. Rademacher wild-cluster-bootstrap p (999 replications): β_1 in M1, $p = 0.001$; β_3 in M2, $p = 0.001$. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (economy-clustered).

5.2 The pre-registered moderation test (H-000500)

The moderation prediction is **not supported, with the interaction carrying the opposite sign**. In M2 the interaction of digital-payment activity with lower 2019 credit-information coverage is **-1.975 percentage points** per one economy-level standard deviation (economy-clustered SE 0.474 pp; $p \approx 6.8 \times 10^{-5}$; 95% CI -2.916 to -1.033 pp; wild-cluster-bootstrap $p = 0.001$). The digital-payment/formal-borrowing association is therefore *weaker*, not stronger, in economies with thinner 2019 coverage — the reverse of the credit-information-substitution prediction. The below-median contrast (M2b) agrees: the association is -3.958 pp smaller in economies below the coverage median (SE 0.970 pp; $p \approx 9.3 \times 10^{-5}$).

The observed clustered precision implies an approximate conventional 80%-power minimum detectable interaction of 1.33 pp per SD at a two-sided 5% level, and the 95% interval is entirely negative, so within this specification the data reject not only the registered positive prediction but any non-negative interaction. This is a model-based precision statement, not evidence that the true structural effect is negative.

Under the frozen decision rule this places H-000500 in the *not supported* branch. We do not relabel the reverse pattern as a confirmation of a different hypothesis; it was not pre-registered.

5.3 Robustness of the registered coverage result

For the registered coverage moderator, the negative interaction persists across the pre-registered weighting, sample and control-set branches and the adversarial specifications (Table 4). The alternative depth and legal-rights proxies do not yield statistically significant interactions and therefore do not generalise the reversal to other institutional constructs.

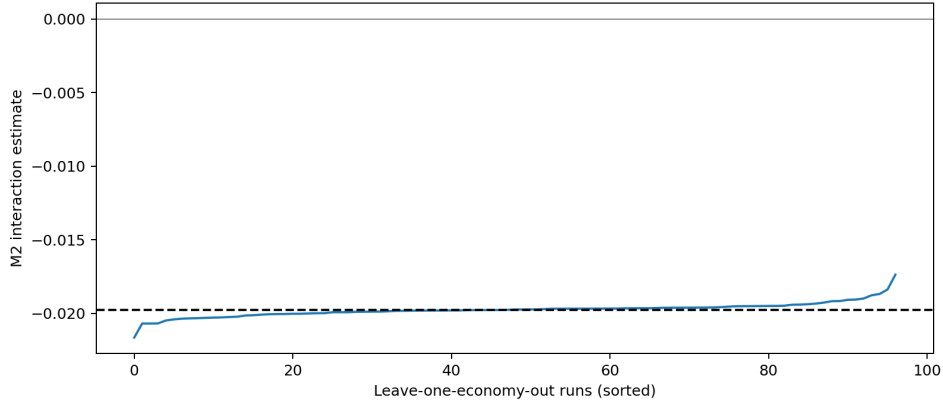


Figure 2: Leave-one-economy-out estimates of the M2 interaction β_3 . Each point drops a single economy and re-estimates; the dashed line is the full-sample estimate and the grey line marks zero. All 97 fits are negative.

Dummy-variable logit and probit give negative interaction average marginal effects (approximately -1.35 and -1.36 pp from the primary run; the independent reproduction obtains -1.50 and -1.47 pp, the same sign with a larger magnitude); the sign, not the precise magnitude, is the informative quantity from these non-linear models. The specifications that do not reach significance are those that also lose precision by construction: the account-holders-only restriction

Table 4: The M2 interaction β_3 across specifications (percentage points per SD).

Specification	β_3	p	N	G	Note
M2 baseline	-1.975	<0.0001	100,560	97	
Unweighted	-2.129	<0.0001	100,560	97	
Population weights	-2.783	0.00046	100,560	97	
Drop 6 high-income economies	-2.050	<0.0001	94,682	91	z re-computed
+ country controls $\times$ digital payment	-2.354	0.003	100,560	97	
+ internet-use control	-2.048	<0.0001	100,560	97	
+ employment control	-1.896	0.00017	96,925	96	
Economies with $N \geq 500$	-1.975	<0.0001	100,560	97	
Moderator = depth-of-information proxy	-0.525	0.182	100,560	97	same sign, imprecise
Moderator = legal-rights proxy	0.705	0.236	100,560	97	wrong sign, imprecise
Restrict to account holders	-0.803	0.218	55,429	97	imprecise
Drop zero-coverage economies	-2.121	<0.0001	99,059	95	
Drop lowest coverage quartile	-2.880	<0.0001	76,108	72	
Drop China and Saudi Arabia; re-standardize	-1.955	0.00010	95,909	95	indep. reproduced
+ control $\times$ moderator (post-registration)	-1.545	0.00091	100,560	97	post-reg., dual impl.
Leave out each of 7 world regions	-1.412 ... -2.187	all negative			
Leave out each of 97 economies	-1.737 ... -2.164	all negative; 0 sign reversals			
Economy-block bootstrap (999)	-1.975	95% percentile CI [-2.91, -1.05]			
Moderator sign reversed (algebra check)	1.975	exact negation of the baseline			

Notes. All specifications are M2-form (percentage points per one economy-level SD of the negative coverage z -score), economy-clustered (CR1). All rows were reproduced by the independent cold implementation CLAUDE-S5R-20260910-001; the China/Saudi-Arabia row was additionally reproduced by CLAUDE-S5R-DB-INTEGRITY-20260910-001 ($|\Delta| = 5 \times 10^{-14}$) and the control-by-moderator row by CODEX-S5-CTRLXMOD-REPRO-20260910-001 ($|\Delta| = 3 \times 10^{-13}$). The control-by-moderator row is a post-registration check outside the frozen quantity set. Dummy-variable logit and probit interaction average marginal effects are -1.348 and -1.358 percentage points (same sign).

and the two alternative moderator constructs. Of the latter, the depth-of-credit-information proxy keeps the negative sign (-0.525 pp); only the legal-rights proxy turns positive (+0.705 pp), and it is far from significant ($p \approx 0.24$) and measures strength of legal rights rather than coverage. No coverage-based specification reverses toward the hypothesised sign.

One threat the frozen plan does not address is composition: M2 interacts only digital-payment activity and account ownership with the moderator, so if the covariate mix of digital-payment users varies with country coverage the interaction could be an artifact. As a post-registration check, outside the frozen quantity set, we add each control interacted with `lowcov2019_z` to M2. The interaction attenuates to -1.545 pp per SD (economy-clustered SE 0.451 pp; $p \approx 0.00091$; 95% CI -2.44 to -0.65) — about a fifth of the baseline magnitude is absorbed once the control slopes are allowed to vary with coverage, consistent with a compositional contribution — but it keeps its sign and significance. The reversal is therefore not a compositional artifact of the control set, though composition plausibly accounts for part of the gradient. A separate implementation reproduces this result within numerical tolerance. The within- R^2 rises only from 0.0481 to 0.0491 (a single-implementation diagnostic whose baseline value matches the frozen Stage-2 run exactly); we report this transparent coefficient-stability comparison instead of treating a scalar selection bound as identification for a cross-level interaction in a binary-outcome fixed-effects model.

5.4 Mechanism and placebo

The registered receive-versus-make mechanism (M3) does not behave as the substitution reading predicts (Table 5). In M3a the making-payment coefficient (5.172 pp) exceeds the receiving-payment coefficient (4.080 pp), contrary to the prediction that incoming, lender-visible payments matter more. In M3b the receive interaction is essentially zero (-0.269 pp, $p \approx 0.587$) while the make interaction is -2.021 pp ($p \approx 6 \times 10^{-5}$). The informal-borrowing placebo (M4) also fails: the interaction for borrowing from family or friends is **positive** (1.736 pp, $p \approx 0.005$; Benjamini–Hochberg $q \approx 0.010$; 100,428 adults), where the substitution reading predicts it to be approximately zero (Table 6). Taken together, the secondary results give no support to a credit-information-substitution interpretation even though some of the individual coefficients are statistically distinguishable from zero.

Table 5: Mechanism split: receiving versus making digital payments.

	Coefficient	(CR1 SE)	p
Receiving a digital payment (M3a)	4.080***	(0.578)	<0.0001
Making a digital payment (M3a)	5.172***	(0.544)	<0.0001
Receiving $\times$ lower 2019 coverage (M3b)	-0.269	(0.494)	0.587
Making $\times$ lower 2019 coverage (M3b)	-2.021***	(0.480)	<0.0001
Observations		90,118	
Clusters (economies)		89	

Notes. Mechanism sub-sample (non-missing receive and make indicators). M3a and M3b carry the full frozen control set and economy fixed effects. Percentage points; economy-clustered (CR1) standard errors. The substitution reading predicts the receiving interaction to be the more negative; it is instead near zero.

Table 6: Secondary outcomes and the informal-borrowing placebo.

Outcome	β_1	β_3	$p(\beta_3)$	BH $q(\beta_3)$	N
Mobile-loan-app use	7.375***	1.097*	0.086	0.086	100,452
Emergency loan from a formal source	0.718**	-0.883**	0.022	0.029	97,540
Informal borrowing (placebo)	10.662***	1.736***	0.005	0.010	100,428

Notes. Each row is one M2-form regression on the named outcome, full control set and economy fixed effects, percentage points, economy-clustered (CR1). Benjamini–Hochberg q -values are computed over the frozen nine-hypothesis secondary family. The substitution reading predicts the informal-borrowing β_3 near zero; it is positive and survives the correction.

5.5 Independent reproduction

The primary and robustness results were produced by one implementation and then reproduced by a separate cold implementation written from the frozen contracts alone. Of 44 compared quantities, 41 match to an absolute difference below 5×10^{-13} (largest 4.8×10^{-13}), including M1, the M2 interaction, M2b, the mechanism split, every secondary outcome, the full robustness grid, and every adversarial check. The three that differ belong to two non-primary diagnostics — the M2c joint-test statistic form, and the logit and probit interaction marginal effects — and differ in magnitude but not in sign or conclusion. The reproduction verdict is PASS. The

post-registration China/Saudi-Arabia integrity sensitivity was likewise reproduced by a separate implementation to an absolute difference of 5×10^{-14} .

6 Vietnam in the cross-country distribution

This section is descriptive; it makes no claim about a Vietnam-specific effect. Vietnam’s 2019 credit-information coverage is 59.4% (public registry 59.4%, private bureau 20.6%), above the 97-economy median of 40.4% — Vietnam is a *higher-coverage* economy on this measure, with a maximum-score depth-of-credit-information index of 8. Evaluated at Vietnam’s value of the moderator, the fitted M2 slope of formal borrowing on digital-payment activity is approximately +5.07 pp (95% CI 3.87–6.27 pp; delta method). This is the cross-country pattern read at Vietnam’s 2019 coverage level, not an estimate of what digital payments do in Vietnam. Both the point estimate and the interval are now persisted (CLAUDE-S5R-M7-VNM-CI-20260910-001); the point estimate matches the primary run to twelve decimal places, and the interval is a descriptive diagnostic (M7) outside the frozen quantity set.

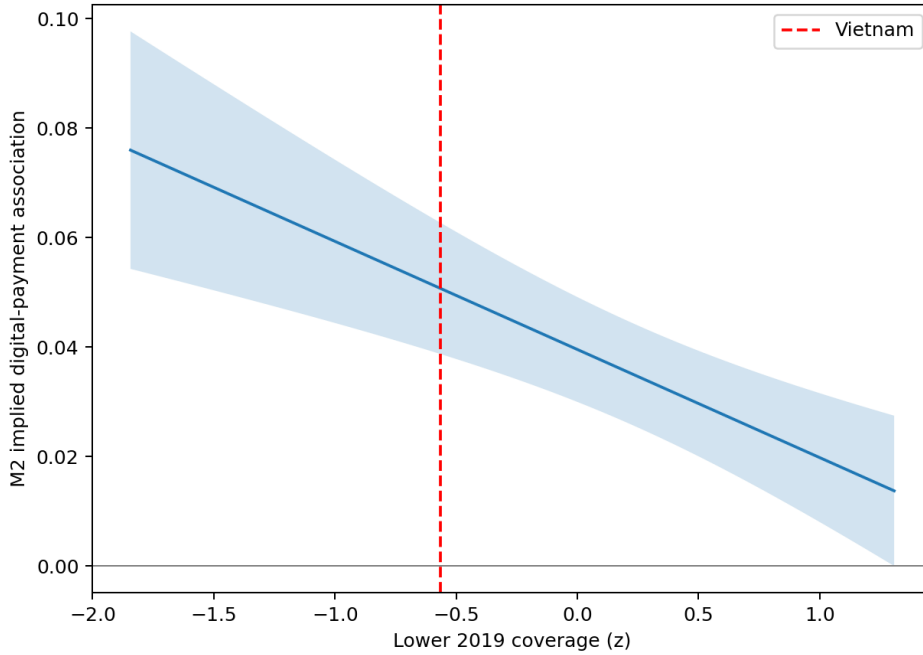


Figure 3: M2-implied digital-payment/formal-borrowing association across the range of the 2019 coverage moderator (higher x = thinner coverage), with a pointwise 95% band. Vietnam’s position is marked. The slope of this line is the pre-registered interaction $\beta_3 = -1.975$ pp per SD.

7 Discussion

Why might the association be weaker, not stronger, where coverage is thin? Three explanations are consistent with the evidence and with the design’s limits.

A coverage floor. In the 24 economies with 2019 coverage below 10%, credit-information systems are close to absent for everyone; there is little cross-sectional room for a digital-payment

record to “substitute” at the margin, and formal borrowing in these economies is low and concentrated among a small, selected group. Dropping the zero-coverage economies and the lowest coverage quartile does not remove the negative interaction, which argues against this being the whole story, but a non-linear coverage relationship remains plausible.

The account-access channel. Where formal financial infrastructure is thin, holding an account and using digital payments may principally mark inclusion in the formal system at all, rather than an incremental, lender-legible signal on top of an existing credit file. The account-holders-only restriction shrinks the interaction toward zero and renders it imprecise, which is consistent with composition — including non-account holders — accounting for part of the gradient. The fully interacted-control check also shows partial attenuation. These clues make the account-access channel a plausible interpretation, but neither identifies it: the restriction is imprecise and conditions on a possible collider, while coefficient attenuation can have several causes. We therefore treat this as a hypothesis for future data, not the paper’s mechanism finding.

Supply-side heterogeneity. Lenders in thin-coverage economies may not have the systems to act on a digital-payment history even when it exists; the firm-level result may reflect relationship-lending technologies that individual retail lending in these settings lacks. We cannot test this with Findex.

Reconciling with Galilea et al. (2026). The unit differs (firms versus individuals), the outcome differs (self-reported credit constraints versus realised formal borrowing), the moderator differs (depth index versus coverage), and the lending technologies differ. A contrary individual-level cross-section does not contradict their firm-level finding; it bounds its external validity.

Policy boundary. The rejected moderation prediction does not imply that digital payments fail to improve access, that credit reporting is unimportant, or that payment data cannot aid underwriting. It says only that this 2024 individual cross-section does not show a larger conditional payment–borrowing association in economies with thinner archived 2019 coverage.

Residual limitations that must survive typesetting. The following limitations are carried verbatim from the independent adversarial review and are not weakened anywhere in this paper:

- The design is a single cross-section; reverse causality and selection remain.
- The cross-level interaction is identified by 97 economy values of an archived 2019 moderator, not by 100,560 independent country observations.
- The Doing Business checks establish agreement with archived profiles/live World Bank data, not validation against a fully independent measurement system. General non-classical measurement error remains possible.
- The depth and legal-rights proxies do not significantly reproduce the coverage result.
- The C5 fully interacted model is post-registration. Its coefficient is dual-implementation; its within- R^2 diagnostic is persisted but single-implementation.

- The MDE is a retrospective precision summary using the observed clustered SE, not an ex ante design guarantee or proof of a negative structural effect.
- The account-access channel is a plausible hypothesis only; it is not identified.
- `search_adequacy` remains PARTIAL, so the paper cannot claim to be the first study.

Additional design-level caveats: `account_fin` is conditioned on but is a possible common consequence, so the account-holders-only check is a sensitivity analysis, not a fix; the private-credit robustness control has heterogeneous vintages; and the Global Findex provenance is verified-reconstructed, not original-download verified. None of these is resolved by the design, and all are why the language stays associational.

8 Conclusion

In a pre-registered individual-level test across 97 economies, digital-payment activity is positively associated with formal borrowing, but the credit-information-substitution prediction — that the association is larger where credit-information coverage is thinner — is not supported. For the registered coverage moderator, the interaction has the opposite sign across weighting, sample and control-set variants and all leave-one-economy-out fits; the two alternative institutional proxies are imprecise. The registered mechanism and placebo tests give no support to the substitution reading. We read this as an external-population check that does not reproduce the firm-level pattern at the individual level in this cross-section. Basic formal-system access is one plausible explanation, but the present design cannot identify it. The value of the exercise is in its transparency: a frozen plan, an independently reproduced result, and a contrary finding reported as such.

Reproducibility statement

Raw inputs, code and results are under `projects/study_05_findex_crosscountry/`. The frozen pre-registration and hypothesis file were hash-committed before estimation. Primary estimation and independent reproduction each have a fixed seed and a recorded input/output SHA-256; every table and in-text statistic in this manuscript is generated by `code/study_05_crosscountry/tables/build_tables.py` from those immutable result files. Input SHA-256: Findex microdata `ca307a0c...f90fce`; 2019 credit-information snapshot `ef0c67f9...bd06c`.

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